

Research objectives determine the information needed, and it must follow a hierarchical order:

1. Assess/Analyze
 2. Design
 3. Develop
 4. Test
 5. Implement/Deploy
 6. Evaluate/Measure
1. **Assess** – understand the problem and gather requirements.
 2. **Design** – plan the system architecture, database, and interface.
 3. **Develop** – build the software.
 4. **Test (Pilot Testing/Use Case Testing)** – verify that the software works correctly (often included within development or evaluation).
 5. **Implement/Deploy** – release the system for intended users, if applicable.

1. Pilot Testing - Pilot testing is a small-scale trial implementation of the completed system with a limited number of actual users to determine whether it is ready for full deployment.

- 1 owner
- 2 pharmacists
- 1 cashier

Researchers observe:

- Were there unexpected errors?
- Did users understand the workflow?
- Were reports generated correctly?
- Was training sufficient?

Pilot testing answers: "Is the whole system ready for actual use?"

2. Use Case Testing - Use case testing verifies that each functional requirement or feature works exactly as specified. It is based directly on the system's use cases.

Example Use Case: Login

Expected behavior:

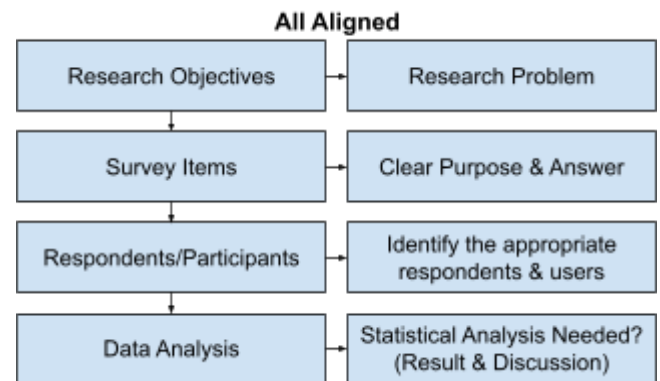
- Enter a valid username
- Enter valid password
- Click Login

Expected result: ✓ Dashboard appears.

Use Case: Add Product

Steps:

1. Click Add Product
2. Fill in product details



6. **Evaluate** – assess the software's quality, usability, functionality, security, and performance using a framework such as ISO/IEC 25010.

Example: Suppose you developed a Pharmacy Management System.

Instead of immediately using it throughout the pharmacy, you allow it to be used for one week.:

3. Save

Expected result: ✓ Product appears in inventory.

Use Case: Generate Sales Report

Expected result: ✓ PDF report is generated.

Use case testing answers: "Does every feature work correctly?"

3. User Testing (Usability Testing) - User testing evaluates how real users interact with the system, focusing on ease of use, efficiency, and satisfaction. It does not primarily check whether features work—that is already covered by functional or use case testing.

Example: Researchers ask 20 users to perform tasks:

- Login
- Create an order
- Print receipt
- Search products

Afterwards, they complete a questionnaire:

- Was the interface easy to understand?
- Were buttons easy to locate?
- Was navigation intuitive?
- Were instructions clear?
- Overall satisfaction?

This often uses:

- System Usability Scale (SUS)
- ISO/IEC 25010 Usability
- User Experience Questionnaire (UEQ)

User testing answers: "Can users easily use the system?"

Which One Should Be Used in a BSIT Capstone?

A typical testing flow is:

1. **Unit Testing** – Individual methods or components are tested.
2. **Integration Testing** – Modules are tested together.
3. **System Testing** – The complete system is tested as a whole.
4. **Use Case Testing (Functional Testing)** – Verify each requirement and feature against expected outcomes.
5. **User Testing (Usability Testing)** – Observe representative users and collect feedback on ease of use and user experience.
6. **Pilot Testing** (optional but recommended if the client will actually use the system) – Conduct a limited

real-world trial before full implementation.

7. **System Evaluation** – Evaluate the overall quality of the software, commonly using ISO/IEC 25010 (e.g., functional suitability, usability, reliability, security, performance efficiency, compatibility, maintainability, and portability).

Which is more important?

For most undergraduate capstone projects:

- **Use Case Testing** is essential because it demonstrates that all required functions work.
- **User Testing** is also highly valuable because it provides evidence that the system is usable by its intended users.
- **Pilot Testing** is appropriate when the client is willing to use the system in an actual operational setting before full deployment. If the system remains a prototype or is not yet adopted by the organization, pilot testing may not be practical or necessary.

Aspect	Pilot Testing	Use Case Testing	User Testing (Usability)
Purpose	Validate the entire system in a real or near-real environment before full implementation	Verify that each functional requirement or use case works correctly	Evaluate how easy the system is for users to learn and use
Focus	Overall readiness	Functional correctness	User experience
Who performs it?	Selected end users and researchers	Developers, testers, QA, or researchers	Representative end users
What is tested?	Complete workflow and system performance	Individual features and scenarios	Ease of navigation, interface, satisfaction
When?	After system testing, before deployment	During testing phase	After functional testing or during evaluation
Output	Identifies real-world issues before rollout	Pass/fail of each use case	User feedback and usability scores

Pre-assessment instruments can be used:

1. Interview (Documented - MOM)
2. Observation (Problem/Lacking & Beneficiary)
3. Questionnaire

Requirements:

1. Documented pictures and signed documents (request, MOM) of both the researcher, client, and adviser, if needed
2. Accomplishment report & survey
3. Validation Forms
4. Request letter for evaluators and approved letter
5. Survey Questionnaire w/ objectives of study

Assessment/Observations:

1. Current/outside workflow
 2. Existing tools/procedures/systems
 3. Time required to complete the task
 4. Difficulties encountered w/ manual
 5. Desired system features
 6. If ready to adopt the system
- System and Documentation must be **defensible** (valid & validated)
 - Validated survey questionnaire (different evaluations for different purposes based on the study or system)

Good Survey Questionnaire:

1. Aligned w/ research objectives, which is aligned with the research problem
2. Must follow and be aligned w/ SDLC stages
3. Must have the information required by researchers
4. Evaluators/Respondents must have qualifications & experience
5. Statistical Analysis (Identify statistical evidence, statistical treatment; every item must have clear options and use tools)
6. ISO 25010 makes sure that it fits/tailored to its own system, is not generalized, and passes 9 characteristics and their subtopics. Can't use ISO 25010 if the system is incomplete; monitor timeline. IT practitioners (20-30) & experts can evaluate.
7. Can have 1 survey that actual users and IT experts can understand in layman's/simple terms; researchers must be present in the evaluation to guide.
8. Measures one idea at a time, focusing on the evaluation of the system
9. Use clean and familiar language (direct concept)
10. Avoid words such as always and never; technical noise. Must have easy wording only.

Survey Questions Examples: Use 4 Likert-Scale

1. The steps to complete the task are easy and understandable.
2. I can use the system without technical knowledge.
3. I intend to use the system when it becomes available.
4. I recommend using the system.
5. I plan to use the system regularly in performing tasks. (employees adapt)

Validation Planning

1. Validators of survey questionnaire: **Research Method Experts, IT Experts/Practitioners, Domain/Subject Experts, Language Master(Grammarian), Statistician**

2. Must be aligned w/ objectives, appropriate construction of order & categorization, no duplicated statements.
3. Suitability of statement procedure.
4. IT Experts still need to review content, system terminology, appropriateness of technical items, tailored to the system, & alignment w/ ISO 25010.
5. Must be ready to present to validators/IT experts.
6. Subject master/Domain expert operation measures accurateness & completeness of indicators.
7. Language master or grammarian needed.
8. Must not have a capstone advisor and panelist to evaluate.
9. At least 3-5 validators for validity and reliability.
10. Need a Statistician to provide guidance w/ statistical techniques and to compute validity tests.
11. Relevance might have revisions
12. Must create a validation form before the letter of request for validators
13. Must bring an approved letter of request, a validation form, a survey questionnaire, and the objectives of the study.
14. Part 1 - Demographic Profile (Background Qualifications), Part 2 - System Evaluation, Part 3 - Comments & Suggestions

Ex:

A letter addressed to the participant, capstone study w/ title, will humbly ask to evaluate our system, and part of the letter will use the Data Privacy Act.

Technology Acceptance Model (TAM):

1. Must have 6 months of usability
2. Final deployment can't keep up with the final period for results submission.
3. Measures user perception and acceptance.
4. Must do before pilot testing w/ 100% use case testing, which is a minimum of 1 year.
5. Each task/process must be completed in 1 day for certain features, summary report generation
6. Must have actual data, process, and report generation
7. Need in TAM perceived usefulness included, which operations/tasks/quality of work/responsibility improved. Users must assess themselves.
8. Behavioral intention to use (willingness), not just pilot testing, but also in the future.
9. If panelists required or included TAM in findings/commented (Matrix), then they must include.
10. Focuses on pre-assessment, planning, existing process/workflow, usefulness & ease/eagerness, or intention to use (beneficial)

Situation & Exposure by TAM:

1. Measures the level of acceptance & usability
2. Simple information (Booking, Tracking, Recording) 1-2 weeks exposure
3. 1 complete operational cycle (Pre-deployment TAM all possible users)
4. Sustained & continuous usage (1-3 months sustainability)

Use Case Testing:

1. Must be 100% pass w/ 0 fails.
2. All processes and features must be presented w/ outputs or outcomes

3. If the use case testing passes, it must already be evaluated
4. Raw data needed
5. Log in to the last process

SOFTWARE PRODUCT QUALITY								
FUNCTIONAL SUITABILITY	PERFORMANCE EFFICIENCY	COMPATIBILITY	INTERACTION CAPABILITY	RELIABILITY	SECURITY	MAINTAINABILITY	FLEXIBILITY	SAFETY
FUNCTIONAL COMPLETENESS	TIME BEHAVIOUR	CO-EXISTENCE	APPROPRIATENESS RECOGNIZABILITY	FAULTLESSNESS	CONFIDENTIALITY	MODULARITY	ADAPTABILITY	OPERATIONAL CONSTRAINT
FUNCTIONAL CORRECTNESS	RESOURCE UTILIZATION	INTEROPERABILITY	LEARNABILITY	AVAILABILITY	INTEGRITY	REUSABILITY	SCALABILITY	RISK IDENTIFICATION
FUNCTIONAL APPROPRIATENESS	CAPACITY		OPERABILITY	FAULT TOLERANCE	NON-REPUDIATION	ANALYSABILITY	INSTALLABILITY	FAIL SAFE
			USER ERROR PROTECTION	RECOVERABILITY	ACCOUNTABILITY	MODIFIABILITY	REPLACEABILITY	HAZARD WARNING
			USER ENGAGEMENT		AUTHENTICITY	TESTABILITY		SAFE INTEGRATION
			INCLUSIVITY		RESISTANCE			
			USER ASSISTANCE					
			SELF-DESCRIPTIVENESS					

ISO 25010:

1. Must evaluate 9 characteristics and their sub-characteristics:

a. Functional Suitability

- i. Functional Completeness
- ii. Functional Correctness
- iii. Functional Appropriateness

b. Performance Efficiency

- i. Time Behaviour
- ii. Resource Utilization
- iii. Capacity

c. Compatibility

- i. Co-existence
- ii. Interoperability

d. Interaction Capability

- i. Appropriateness Recognizability
- ii. Learnability
- iii. Operability
- iv. User Error Protection
- v. User Engagement
- vi. Inclusivity
- vii. User Assistance
- viii. Self-Descriptiveness

e. Reliability

- i. Faultlessness
- ii. Availability
- iii. Fault Tolerance
- iv. Recoverability

f. Security

- i. Confidentiality
- ii. Integrity
- iii. Non-repudiation
- iv. Accountability

v. Authenticity

vi. Resistance

g. Maintainability

i. Modularity

ii. Reusability

iii. Analysability

iv. Modifiability

v. Testability

h. Flexibility

i. Adaptability

ii. Scalability

iii. Installability

iv. Replaceability

i. Safety

i. Operational Constraint

ii. Risk Identification

iii. Fail Safe

iv. Hazard Warning

v. Safe Integration

2. Must justify if not used in the methodology
3. Findings based on objectives; must use and require all.
4. Balance the adviser's and the panel's opinions.
5. IT Experts: 10 evaluators
6. Actual qualified users can't evaluate maintainability, so write to understand.
7. Web developers, data analysts with at least 5 years of experience w/ more than 1 project. (IT Experts)
8. Need validators/evaluators related to the system; if web-based, then web developers; if it involves analytics, then data analysts.
9. Measures if the system has good quality.

Deployment:

1. Letter of approval noted by Capstone & academic adviser.

Notes:

1. For quantitative data, ask a statistician
2. Uploaded in the LMS guide for reliability, validity